F12 4th CAR

Lecture 21: Section 3.2
Logarithmic Functions

Logarithmic function with base a

Properties of logarithms

Inverse properties

One-to-one properties

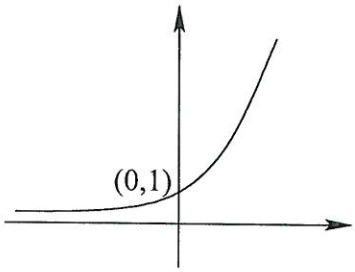
Graphs of logarithmic functions

Natural logarithmic function

Common logarithmic function

$y = a^x$

Recall the graph of exponential function $f(x) = a^x$, $a > 1$:



Since $f(x)$ is one-to-one, it has an inverse function.

Def. The **logarithmic function with base** a, where $x > 0$, $a > 0$ and $a \neq 1$ is written $f(x) = \log_a x$ and is defined by the relationship

$$y = \log_a x \quad \text{if and only if} \quad a^y = x$$

ex. Write in exponential form:

1) $\log_3\left(\frac{1}{9}\right) = -2$ $\boxed{3^{-2} = \frac{1}{9}}$

2) $\log_e(3x + 1) = 2$

$\boxed{e^2 = 3x+1}$

ex. Write in logarithmic form: $4^{1.5} = 8$

$\log_4 8 = 1.5$

ex. Evaluate:

$6 =$ 1) $\log_2 64 = y \rightarrow 2^y = 64$ $\quad 2^6 = 64$ thus $y = 6$

$0 =$ 2) $\log_3 1 = y \rightarrow 3^y = 1$ $\quad 3^0 = 1$ thus $y = 0$

$1 =$ 3) $\log_3 3 = y \rightarrow 3^y = 3$ $\quad 3^1 = 3$ thus $y = 1$

$-3 =$ 4) $\log_{10}\left(\frac{1}{1000}\right) = y \rightarrow 10^y = \frac{1}{1000}$, $10^y = 10^{-3}$ thus $y = -3$

$\frac{1}{2} =$ 5) $\log_{16} 4 = y \rightarrow 16^y = 4$ $\quad 16^{1/2} = 4$ thus $y = \frac{1}{2}$

D.N.E 6) $\log_2(-1) = y \rightarrow 2^y = -1$ no solution

D.N.E. = 7) $\log_2 0 = y \rightarrow 2^y = 0$ no solution

Properties of Logarithms

1. Recall: If $y = \log_a x$, then $x = a^y$

   That is, $x > 0$. thus D: $(0, \infty)$

* 2. $\log_a 1 = 0$

* 3. $\log_a a = 1$

4. **Inverse Properties:** $a^y = a^x$ thus $x = y$

* $\boxed{\cancel{\log_a} a^x = x}$ for all real number x

* $a^{\log_a x} = x$ for $x > 0$

$(\log_5 5)^4 = \log_5 5^{-1} \neq \log_5(5^{-1})$

5. **One-to-One Properties:**

   If $\log_a x = \log_a y$, then $x = y$.

ex. Evaluate:

1) $2^{\log_2 3\pi} = \boxed{3\pi}$

2) $\log_5 \frac{1}{5} = \log_5(5^{-1}) = \boxed{-1}$

ex. Solve: $\log_2(x-3) = \log_2 9$

$x - 3 = 9$

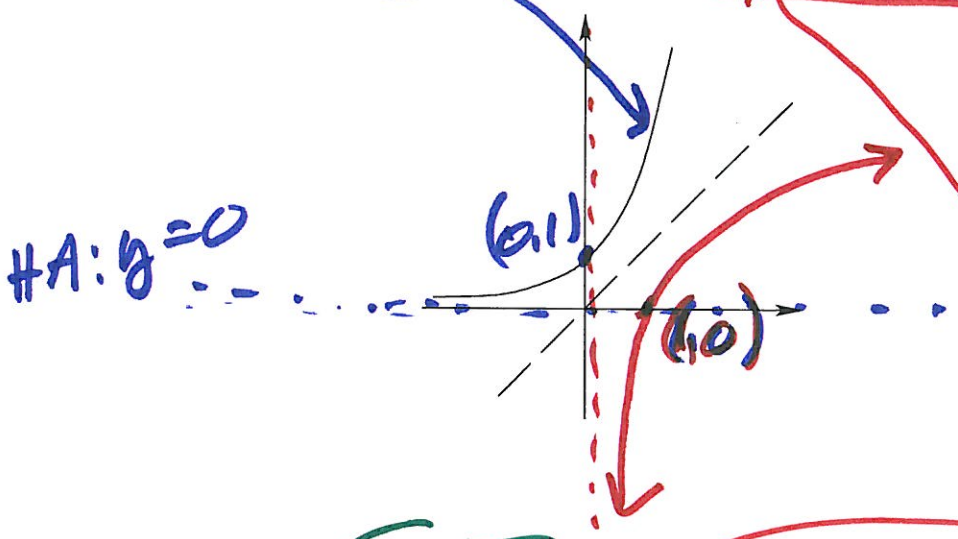
$\boxed{x = 12}$

*Make sure the solution is in the domain.

Graphs of Logarithmic Functions

ex. Sketch $y = 2^x$ and $y = \log_2 x$. $\rightarrow 2^y = x$

VA: $x=0$



memorize basic shape

ex. Sketch $y = \left(\frac{1}{2}\right)^x$ and $y = \log_{\frac{1}{2}} x$ $\rightarrow \left(\frac{1}{2}\right)^y = x$

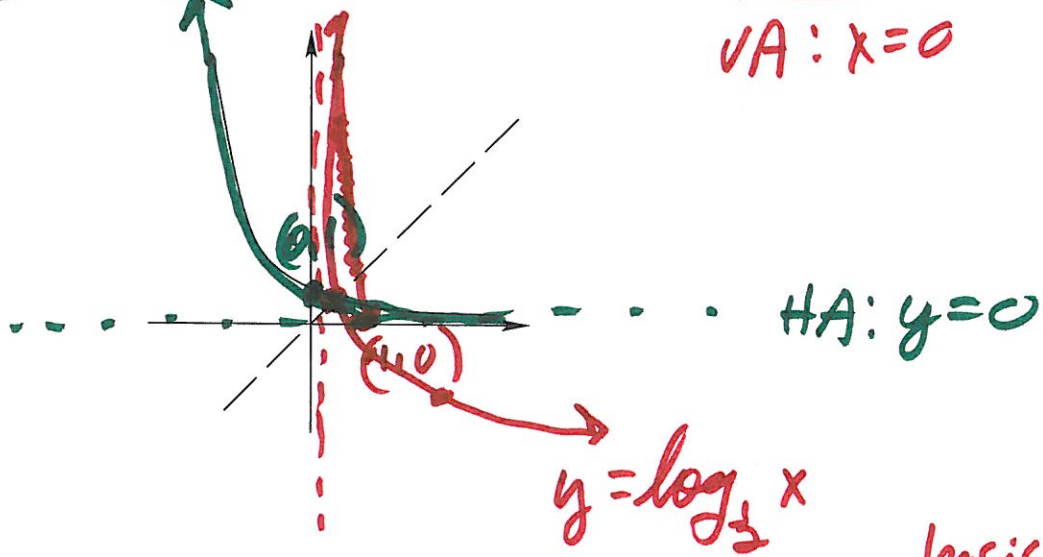
VA: $x=0$

$0<a<1$

In a bind:

$\left(\frac{1}{2}\right)^y = x$

x	y
4	-2
2	-1
1	0
$\frac{1}{2}$	1
$\frac{1}{4}$	2



memorize basic shape.

Properties of the graph of $f(x) = \log_a x$

Compare $f(x) = \log_a x$ and $f^{-1}(x) = a^x$:

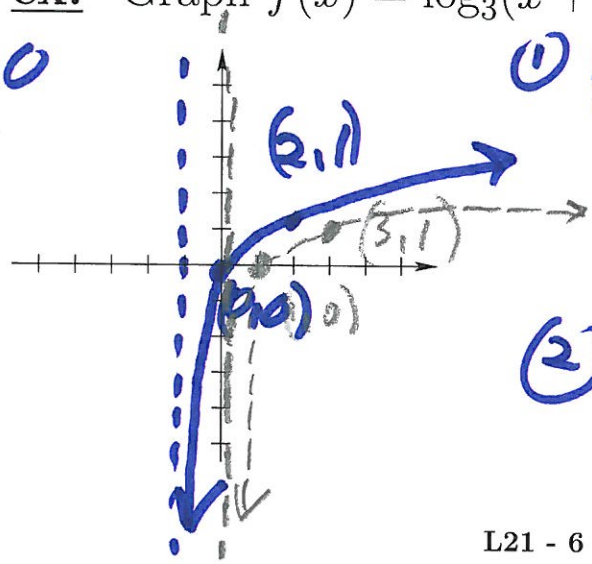
	$f(x) = \log_a x$	$f^{-1}(x) = a^x$
1. Domain:	$(0, \infty)$	$(-\infty, \infty)$
2. Range:	$(-\infty, \infty)$	$(0, \infty)$
3. Intercept:	x-int. $(1,0)$	y-int $(0,1)$
4. Asymptote:	VA: $x=0$	HA: $y=0$
5. increasing if	$a>1$	$a>1$
decreasing if	$0<a<1$	$0<a<1$
6. points on the graph	$(a,1)$, $(\frac{1}{a},-1)$	$(1,a)$, $(-1,\frac{1}{a})$

ex. Graph $f(x) = \log_3(x+1)$

D: $x+1>0$
$x>-1$

R: $(-\infty,\infty)$

VA. $x=-1$



① $y = \log_3 x$
$(3,1)$ $(1,0)$
VA: $x=0$

② $y = \log_3(x+1)$
$\Leftarrow 1$

or $y = \log_3(x+$
$3^y = x+1$
$3^y - 1 = x$

x	y
2/3	-1
0	0
2	1
8	2

The Natural Logarithmic Function base e

$y = \log_e x = \ln x$ if and only if $e^y = x$

Note the following:

$\ln 1 = 0$ $\qquad$ $\ln e = 1$

since $e^0 = 1$ $\qquad$ $e^? = e$

Inverse Properties:

$e^{\ln x} = x$ for $x > 0$

$\ln(e^x) = x$ for all real numbers ✓

One-to-One Property:

If $\ln x = \ln y$, then $x = y$

ex. Evaluate:

D: $2x+3 > 0$

1) $e^{\ln(2x+3)} = 2x+3$ for $x > \frac{-3}{2}$

2) $\ln\left(\frac{1}{e}\right) = \ln_e(e^{-1}) = \boxed{-1}$

ex. Solve: $\ln(x^2 - x) = \ln 6$

$x^2 - x = 6$

$x^2 - x - 6 = 0$

$(x-3)(x+2) = 0$

$\boxed{x = 3,\ x = -2}$ Must check domain.

Plug in: $3^2 - 3 > 0$ ✓

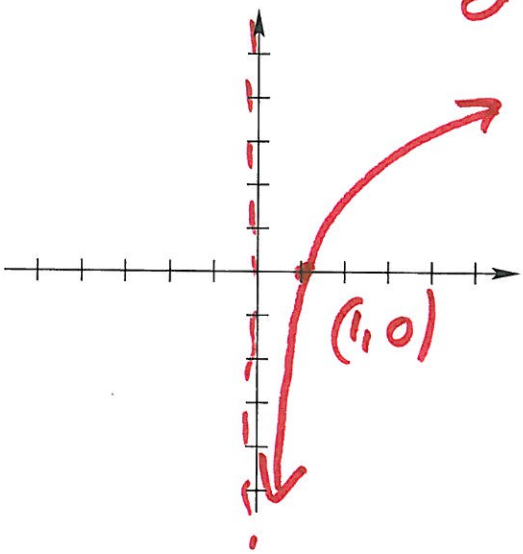
$(-2)^2 - (-2) > 0$ ✓

Remember $2 < e < 3$

$y = \log_e x$

ex. Graph and find the domain and vertical asymptote of $f(x)$:

1) $f(x) = \ln x = \log_e x$



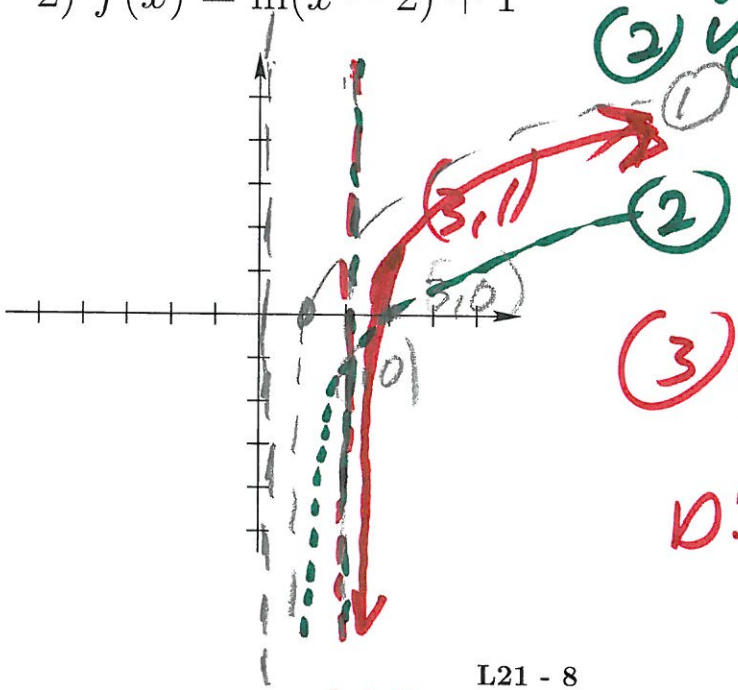
D: $(0, \infty)$

R: $(-\infty, \infty)$

2) $f(x) = \ln(x - 2) + 1$

(1) $y = \ln x$ $(1,0)$

(2) $y = \ln(x-2)$ $\rightarrow 2$



(3) $y = \ln(x-2) + 1$

D: $x - 2 > 0$

$x > 2$

D: $(2, \infty)$

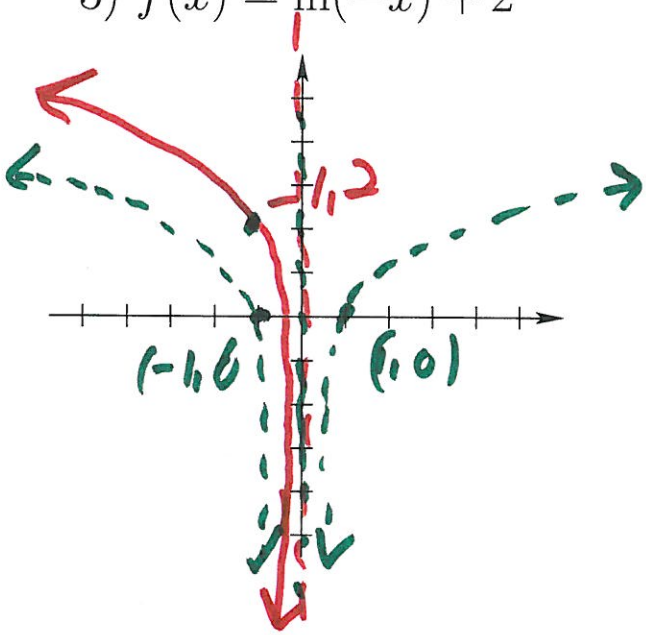
R: $(-\infty, \infty)$

VA: $x = 2$

Note: D: $-x > 0$
$x < 0$

① $y = \ln x$
② $y = \ln(-x)$ refl across the y-axis

3) $f(x) = \ln(-x) + 2$



③ $y = \ln(-x) + 2$
$\uparrow 2$

Common Logarithmic Function

$y = \log_{10} x = \log x$ if and only if $10^y = x$

ex. Evaluate:

$\log 1 = \boxed{0}$

$\log_{10} 10 = \boxed{1}$

$\log 10000 =$

$\log_{10} 10^4 = \boxed{4}$

$\log \frac{1}{\sqrt{10}} = \log_{10} (10)^{-1/2} = \boxed{-\frac{1}{2}}$

Applications

ex. The loudness level of a sound, D, in decibels, is given by

I is variable

$$D = 10 \log \left(\frac{I}{10^{-12}} \right),$$

where I is the intensity of a sound in watts per square meter.

1) Determine the decibel level of a sound with an intensity of 10^{-2} watt per square meter.

$$D = 10 \log \left(\frac{10^{-2}}{10^{-12}} \right) = 10 \log 10^{10} = 10 \cdot 10 = 100$$

$= 10$

100 decibels

$I = 1$

2) Determine the decibel level of a sound with an intensity of 1 watt per square meter.

$$D = 10 \log \frac{1}{10^{-12}} = 10 \log_{10} 10^{12} = 10 \cdot 12 =$$

$= 12$

120 decibels

3) The intensity of a sound in part (2) is 100 times as great as the intensity in part (1). By how much is the decible level increased?

part 2 $I = 1$

part 1 $I = 10^{-2} = \frac{1}{100}$

$$\begin{array}{r} 120 \\ -100 \\ \hline 20 \end{array}$$

20 decibels

Practice.

Determine the domain of each function.

1) $f(x) = \log(x^2 - 2x)$

2) $g(x) = \log\left(\dfrac{3-x}{x+3}\right)$

3) $h(x) = \ln|x-2|$

Answer. 1) $(-\infty, 0) \cup (2, \infty)$ 2) $(-3, 3)$ 3) $(-\infty, 2) \cup (2, \infty)$